Deepfake Text Detection Systems for Arabic and English Using BERT and ELECTRA-Based Models

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Abstract

The rapid advancement of large language models (LLMs) has made AI-generated text increasingly human-like, raising concerns regarding academic integrity, misinformation, and authorship authenticity. Detecting "deepfake text" has therefore become an essential challenge, particularly in multilingual contexts where resources remain limited. This study investigates the effectiveness of transformer-based models in distinguishing human-written from AI-generated text, focusing on Arabic and English fine-tuning while preparing resources for future Arabic detection. Two datasets were collected and preprocessed, and three different transformer models were fine-tuned and evaluated for binary AI-text classification. Experimental results demonstrate clear performance differences between models, with DeBERTa-v3-base achieving the highest accuracy for the English language and AraELECTRA for the Arabic language. The findings highlight both the potential and the limitations of current transformer architectures for deepfake-text detection and provide foundational insights for extending detection systems to multilingual settings, including Arabic.

Keywords—text detection, deepfake, transformer models, BERT model, ELECTRA model.

1 Introduction

The emergence of advanced large language models (LLMs) has fundamentally changed how digital text is produced. Contemporary systems can now generate essays, reports, explanations, and narratives that closely resemble human writing in coherence, fluency, and style. Although these capabilities enable productive uses in education, media, and professional communication, they also introduce serious challenges. AI-generated text can be misappropriated to create misleading information, support academic misconduct, imitate authorship, or automate large-scale content generation that obscures the authenticity of online discourse (2; 1).

As LLMs continue to improve in their mastery of linguistic structure, including languages with complex morphology such as Arabic, the task of distinguishing synthetic text from genuine human writing becomes increasingly difficult. Many existing detection methods depend on shallow linguistic artifacts, perplexity irregularities, or generator-specific patterns, yet these signals rapidly diminish as models evolve or as paraphrasing and rewriting techniques are applied (3). This growing sophistication highlights the need for more reliable and adaptable detection systems that can operate across languages, topics, and writing conditions.

1.1 Problem Statement

The research question of this study is, "How effectively can fine-tuned transformer models distinguish between human-written and AI-generated text?" This study aims to develop a reliable system capable of distinguishing between human-written and AI-generated text by fine-tuning transformer-based models on English datasets and preparing Arabic data for future multilingual extensions. This aligns with recent research highlighting the urgent need for robust and generalizable AI-text detectors that can operate across diverse domains and writing styles (3).

Our objectives for this project are to collect datasets for both Arabic and English and prepare them for deepfake text detection, perform data cleaning to remove noise and inconsistencies, apply text normalization for language standardization, fine-tune a pretrained multilingual model on the labeled dataset, and lastly evaluate the model's performance using standard metrics.

This study aligns with SDG 4: Quality Education, SDG 9: Industry, Innovation, and Infrastructure, and SDG 16: Peace, Justice, and Strong Institutions. Overall, this project contributes to the broader societal need for trustworthy AI systems by addressing the rising challenge of AI-generated text across linguistic and cultural contexts.

2 Prior Literature

2.1 Overview

Research on AI-generated text detection has expanded considerably as large language models (LLMs) continue to advance in fluency and coherence. Existing detection approaches generally fall into three main categories: statistical and perplexity-based methods, embedding-based or contrastive learning methods, and transformer-based classification models. Across these approaches, the literature consistently highlights the difficulty of distinguishing AI-generated text from human writing, especially in multilingual or cross-domain contexts (1; 2; 3).

2.2 Statistical and Perplexity-Based Methods

Early detectors relied on probabilistic measures such as perplexity, token likelihood, and sampling irregularities. Tools such as GLTR and DetectGPT assume that machine-generated text is more predictable than human writing. However, their effectiveness declines sharply with modern instruction-tuned models or paraphrased content, and they perform poorly in multilingual scenarios (1).

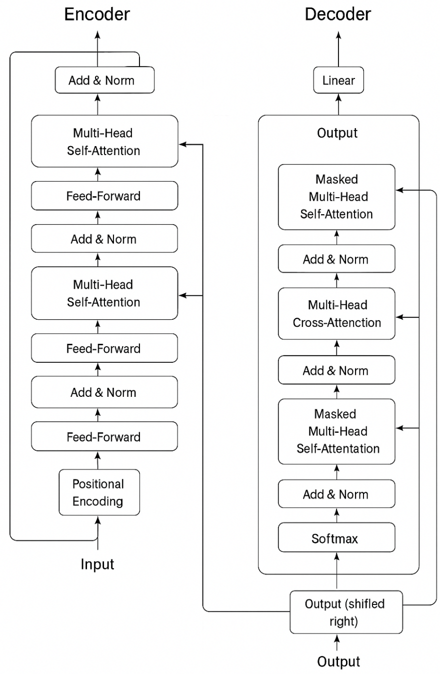
2.3 Contrastive and Representation-Learning Methods

Recent work emphasizes learning discriminative representations that separate AI-generated and human-written text. Models such as Detective employ contrastive learning to magnify embedding differences between human and machine texts (4). Benchmark frameworks such as Genaidetect reveal that many detectors overfit to specific LLM families and exhibit weak robustness to unseen generators (5).

2.4 Transformer Architectures in AI-Text Detection

Transformers have become the dominant architecture in modern NLP since the introduction of the "Attention Is All You Need" model by Vaswani et al. (2017) (8). Their multi-head self-attention mechanism allows them to capture long-range dependencies and contextual relationships more effectively than earlier RNN- or CNN-based architectures.

Because transformers form the backbone of LLMs, they are also widely used in AI-text detection. Fine-tuned transformer encoders learn subtle differences in coherence, semantic depth, structural variation, and stylistic patterns that help distinguish human-written content from AI-generated text. These strengths make transformers highly suitable for multilingual detection tasks, including English and Arabic.



Transrfèrnerchitecture (Vasvani et al.,(2017).

Figure 1: Transformer encoder-decoder architecture

2.5 Summary of Existing Detection Models

Table 1 summarizes the primary models used for AI-text detection in both English and Arabic research, highlighting their strengths, limitations, and the key studies that support their evaluation. These gaps highlight the need for research that systematically compares transformer-based detectors and builds resources suitable for both English and Arabic contexts.

3 Dataset

For this project, two benchmark datasets were selected: ALHD (A Large-Scale and Multigenre Benchmark Dataset for Arabic LLM-Generated Text Detection) and balanced_ai_human_prompts, to develop a reliable multilingual deepfake text detection system.

Table 1: Comparison of AI-Text Detection Models in English and Arabic Literature

Methods Compared	**Dataset Used (EN/AR)**	**Best Performance**	**Key Findings**	**Gaps Identified**	**Refs**
GLTR, DetectGPT (Statistical Baselines)	English datasets	Effective on older LLMs	Highlight predictable token likelihoods; lightweight and fast for screening.	Fail on advanced LLMs; ineffective under paraphrasing; no Arabic evaluation.	(1)
Detective (Contrastive Learning)	English benchmark datasets	Strong semantic separation	Contrastive learning improves separation of human vs. AI representations.	Overfits to specific generators; lacks multilingual evaluation.	(4)
Genaidetect Benchmark	English (multiple LLM families)	High-quality black-box evaluation	Assesses cross-model robustness; identifies failures on unseen generators.	Arabic not included; performance drops sharply with unseen LLMs.	(5)
Transformer Classifiers (BERT, RoBERTa, DeBERTa)	English datasets (essays, news, general text)	High accuracy in controlled settings	Capture contextual and semantic features; state-of-the-art for English AI-text detection.	Degraded performance on short text, noise, and domain shift; limited multilingual testing.	(3)
AraBERT, MARBERT (Arabic Transformers)	Arabic datasets (MSA + dialect corpora)	Best Arabic performance to date	Strong morphological and linguistic modeling for Arabic.	Very few Arabic AI-generated datasets; limited studies on AI-text detection in Arabic.	(9)
KFUPM–JRCAI Dataset Studies	Arabic academic abstracts	Reliable detection within academic writing	Largest Arabic academic AI-generated dataset; supports early Arabic detection research.	Domain-limited (academic only); lacks diverse genres and informal text.	(10)

3.1 ALHD

The ALHD (A Large-Scale and Multigenre Benchmark Dataset for Arabic LLM-Generated Text Detection) is a comprehensive Arabic dataset developed to advance research in detecting AI-generated content. The dataset contains 20,268 Arabic text samples, representing a balanced 10% subset of the full ALHD corpus. It spans a wide range of genres, including news articles, tweets, product and book reviews, health content, and conversational text. The dataset is evenly balanced between human-written and machine-generated samples, with synthetic texts produced by an advanced language model, which is GPT-3.5-T, shown in Figure 2 preventing the model from becoming biased toward one class. This makes ALHD a rich and diverse resource for evaluating Arabic deepfake detection models.

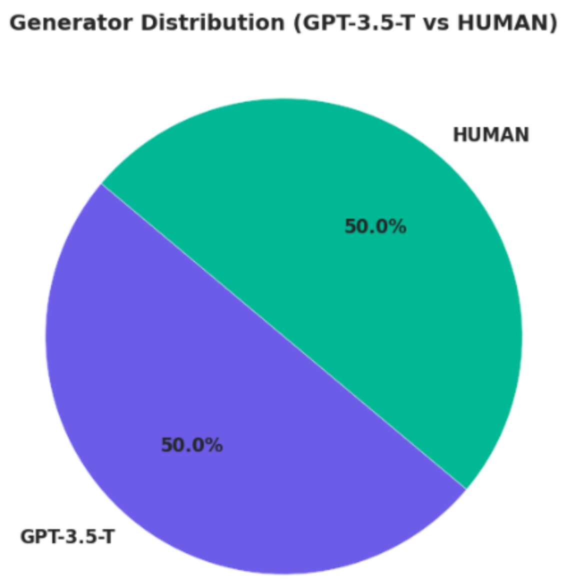


Figure 2: Balanced distribution of HUMAN and GPT-3.5-T generated texts (50% each) in ALHD dataset

For our project, we downloaded the dataset to our Google Drive account and then mounted it. We did basic exploration on the dataset, and we found that it contains 11 columns: 'source', 'document_id', 'text', 'label', 'generator', 'category', 'subcategory', 'token_count', 'text_for_arabert_base', 'text_for_arabert_large', and 'text_for_araelectra'. It has a (20268, 11) shape and no null values.

3.2 balanced_ai_human_prompts

The balanced_ai_human_prompts dataset is an English-language benchmark for evaluating classifiers' ability to distinguish between human-written and AI-generated text. It contains an equal number of human and machine-generated prompts as shown in Figure 3, ensuring a balanced class distribution for training and evaluation. The AI-generated samples were produced using GPT-style models GPT-3 and GPT-3.5, resulting in well-structured text that reflects typical LLM output. Because of its balanced composition and clear labeling, this dataset is well-suited for research on deepfake text detection, model comparison, and general exploration of linguistic differences between human and machine-generated content.

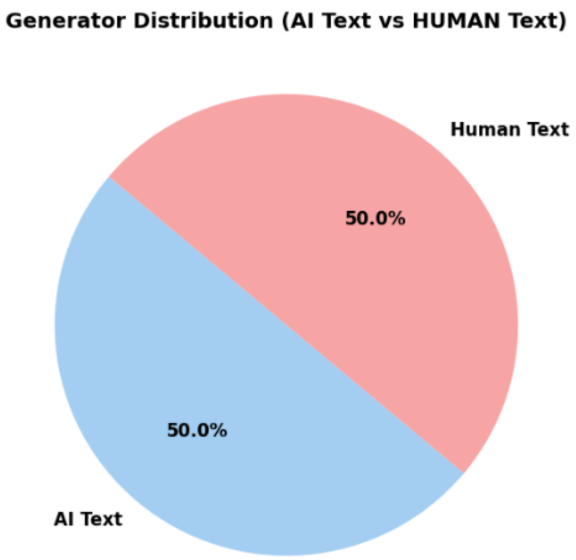


Figure 3: The balanced_ai_human_prompts dataset shows a balanced split between human and AI texts

For this study, the balanced_ai_human_prompts dataset was downloaded to a Google Drive account and then mounted. A basic exploration on the dataset was done, and we found that it contains 2 columns: 'text' and 'generated.' It has a (2750, 2) shape and no null values. Here are some code snippets demonstrating the basic exploration of the dataset.

Together, ALHD and balanced_ai_human_prompts enable cross-lingual evaluation and provide a strong foundation for training and benchmarking deepfake text detection models in both Arabic and English.

4 Methods

4.1 Model Training and Implementation (Arabic)

In this project, multiple pretrained Arabic transformer models (AraBERT-base, AraBERT-large, and AraELECTRA) were fine-tuned using the ALHD dataset to classify text as either human-written or machine-generated. The implementation follows a structured pipeline consisting of preprocessing, tokenization, model training, evaluation, and inference.

4.1.1 Tokenization and Input Preparation

For each model, the corresponding tokenizer was loaded using `AutoTokenizer` and applied to the dataset. Tokenization handled truncation, padding, and sequence length normalization (`max_length = 128`), converting raw text into numerical input IDs suitable for transformer models.

4.1.2 Dataset Construction

The tokenized outputs and labels were wrapped into a custom PyTorch dataset class (`FakeDataset`). This allowed efficient batch processing and seamless integration with HuggingFace's `Trainer` API.

4.1.3 Model Fine-Tuning

Each model was loaded using `AutoModelForSequenceClassification` with a binary classification head (`num_labels = 2`). To reduce memory consumption, gradient checkpointing and FP16 mixed-precision training were enabled.

Training was performed using HuggingFace's `Trainer` with the following configuration:

- Batch size: 8
- Learning rate: 2×10^{-5}
- Number of epochs: 3
- `fp16 = true` for mixed-precision optimization

- `save_strategy = "no"` to avoid disk overflow
- `report_to = "none"` to disable external logging

4.1.4 Model Evaluation

A separate validation and test split were used to compute accuracy and F1-score. A custom `compute_metrics` function applied softmax to model logits and produced prediction labels and probability scores.

4.1.5 Unified Prediction Function

A dedicated inference function (`predict_with_all_models`) allows the system to take any input text and evaluate it across all trained models. For each model, it outputs:

- Predicted label (`REAL` or `FAKE`)
- Probability that the text is machine-generated

This provides real-time prediction capability and enables performance comparison between models.

4.2 Model Training and Implementation (English)

For the English experiments, three pretrained transformer models—DistilBERT, BERT-base, and DeBERTa-v3-base—were fine-tuned on the Balanced_AI_Human_Prompts dataset, which contains an equal number of human-written and machine-generated samples. The implementation followed a structured pipeline consisting of preprocessing, tokenization, dataset construction, model training, evaluation, and inference.

4.2.1 Text Preprocessing

A dedicated preprocessing function was applied to clean the raw English text before tokenization. The preprocessing steps included converting all text to lowercase, removing URLs using regular expressions, and normalizing whitespace. These operations reduced noise and ensured consistent formatting across all input samples.

4.2.2 Train/Validation Split

The dataset was divided into training and validation subsets using an 80–20 stratified split. Stratification ensured that both sets preserved the original balance between human-written and AI-generated text, enabling fair and stable evaluation across all models.

4.2.3 Tokenization and Input Preparation

Each English model was paired with its corresponding pretrained tokenizer, loaded through the HuggingFace `AutoTokenizer`. Tokenization produced input IDs and attention masks while applying padding, truncation, and a maximum sequence length of 256 tokens. The tokenized data and associated labels were wrapped into PyTorch dataset objects to support efficient mini-batching.

4.2.4 Dataset Construction

A custom PyTorch dataset class (`TextDataset`) was used to store the tokenized inputs and their labels. Training and validation data were then loaded using `DataLoader` objects, with a batch size of 16 to enable efficient GPU utilization.

4.2.5 Model Fine-Tuning

Each model was initialized using `AutoModelForSequenceClassification` with a binary classification head (`num_labels = 2`). Fine-tuning was carried out using a manual PyTorch training loop, where the model parameters were updated based on the cross-entropy loss computed for each batch.

The fine-tuning configuration included the following hyperparameters:

- **Batch size:** 16
- **Number of epochs:** 8
- **Learning rate:** 2×10^{-5} (AdamW optimizer)

During training, the model iteratively processed batches of tokenized inputs, computed the loss, performed backpropagation, and updated its parameters. This process enabled each model to specialize in distinguishing between human-written and machine-generated English text.

4.2.6 Model Evaluation

Model evaluation was performed using a custom inference loop that operated in `eval` mode with gradient computation disabled. Predictions were obtained by selecting the most probable class from the model logits. Accuracy and F1-score were calculated using the scikit-learn metrics, providing a consistent basis for comparing the performance of the three English models.

4.2.7 Unified Prediction Function

A unified inference function was implemented to enable real-time evaluation across all three English models. For any input text, the function returns both the predicted class (REAL or FAKE) and the probability that the text is machine-generated. This interface supports rapid comparison and practical deployment.

5 Analysis

5.1 Performance analysis in Arabic dataset

Evaluating different Arabic language models on the task of detecting machine-generated text reveals how each architecture interprets linguistic patterns and handles subtle stylistic differences. Even though all three models were fine-tuned on the same dataset, they approach the problem with different strengths depending on how they were pretrained and the type of representations they learn.

AraBERT-Large, with its deeper architecture, tends to build very rich contextual embeddings, but this depth sometimes works against it in classification tasks where simpler patterns dominate the decision process. The model captures nuanced information but becomes more sensitive to noise and writing-style variations, leading to less consistent predictions.

AraBERT-Base provides a more stable middle ground. Its lighter architecture focuses on essential features without overcomplicating the representation. As a result, it maintains steadier behavior and generalizes well across different writing styles, making it reliable for practical deployment.

AraElectra differs fundamentally from the BERT-based models because of its discriminative pretraining strategy. Instead of predicting masked tokens, it learns to detect replaced tokens, which trains it to notice unnatural or inconsistent phrasing. This characteristic aligns perfectly to distinguish human-written from AI-generated Arabic text, allowing it to capture irregularities that the other models may overlook.

Overall, the comparison highlights that training methodology matters more than model size. While AraBERT-Large offers depth and AraBERT-Base brings balance, AraElectra's unique learning process gives it the strongest capability for this specific detection task.

Aspect	**AraBERT-Large**	**AraBERT-Base**	**AraElectra**
Efficiency	Slowest	Moderate	Fastest
Accuracy	92.97	93.81	96.50
F1-Score	0.9327	0.9399	0.9657
Evaluation Stability	Moderate	High	Very High
Contextual Behavior	Deep but inconsistent	Balanced	Highly sensitive to token-level cues
Generalization	Good	Strong	Strongest

Table 2: Comparison of Arabic transformer models for fake-text detection.

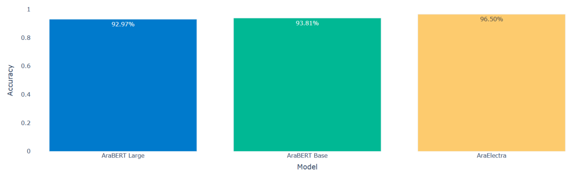


Figure 4: Arabic Models Evaluation

5.2 Performance Analysis (English Models)

The three transformer models—DistilBERT, EngBERT (BERT-base), and DeBERTa-v3-base obtained nearly identical accuracy and F1-scores on the English dataset, indicating that the AI-versus-human detection task is highly separable. However, a deeper examination reveals meaningful behavioral differences between the architectures. DistilBERT, despite its smaller size, learned the core discriminative patterns effectively but showed slightly higher variance during evaluation, suggesting sensitivity to ambiguous or longer samples. BERT-base benefited from its deeper layers and richer contextual embeddings, resulting in smoother convergence and more stable generalization across validation data. DeBERTa-v3-base achieved the most consistent and reliable performance due to its disentangled attention mechanism, which captures long-range dependencies more effectively than standard Transformers. These differences highlight that although all models score similarly, their internal representations, stability, and robustness vary, making each model more suitable for different computational and deployment scenarios.

In conclusion, DistilBERT is the most efficient model, BERT-base offers a strong balance between performance and resources, and DeBERTa-v3-base provides the highest stability and reliability.

Aspect	DistilBERT	EngBERT	DeBERTa-v3-base
Efficiency	Fastest	Moderate	Most Heavy
Accuracy	∼0.9458	∼0.9634	∼0.9789
F1-Score	∼0.9431	∼0.9617	∼0.9792
Stability	Slight Variance	Stable	Most Stable
Semantic Depth	Limited	Strong	Strongest
Generalization	Good	Strong	Strongest

Table 3: Performance comparison of fine-tuned English transformer models.

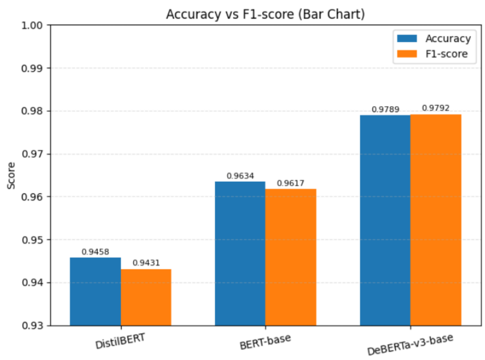


Figure 5: English Models Evaluation

6 Known Project Limitations

This project faced several limitations that may have influenced the models' performance. First, the Arabic experiments were conducted using only the 10% balanced subset of the ALHD dataset, which reduced the amount of training data available and may have limited the models' ability to generalize. Additionally, the dataset contains only formal Modern Standard Arabic, with no representation of regional dialects, which restricts how well the models might perform on real-world, informal, or dialectal text. Training was also constrained by the limited GPU resources available in Google Colab, which imposed restrictions on batch size, sequence length, and training duration. Finally, the classification task was limited to a simple binary setup, human versus AI, which does not capture more complex scenarios such as mixed human–AI writing, partially edited AI text, or identifying the specific model that generated the content.

7 Conclusion

This study examined the effectiveness of transformer-based language models in distinguishing human-written from AI-generated text in both English and Arabic. By evaluating a diverse set of transformer-based models, the work demonstrated that model performance is strongly influenced not only by size but also by pretraining strategy and representational behavior. The English models—DistilBERT, BERT-base, and DeBERTa-v3—achieved consistently strong results, with DeBERTa-v3 providing the most stable and reliable predictions due to its disentangled attention mechanism.

For Arabic, the evaluation revealed a different dynamic. AraBERT-Large and AraBERT-Base performed well, but AraElectra outperformed both models. Its replaced-token pretraining objective made it especially sensitive to stylistic irregularities and token-level inconsistencies commonly found in machine-generated text. This highlights that discriminative training approaches can provide a substantial advantage in fake-text detection, particularly for morphologically rich languages.

Overall, the findings emphasize that AI-generated text detection is increasingly feasible using modern transformer architectures. Cross-lingual experiments further illustrate that model behavior varies across languages, suggesting that architecture selection should be tailored to the linguistic features and available datasets of each target language. The insights gained from this work can support future research in authorship verification, digital forensics, and responsible AI deployment.

8 References

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